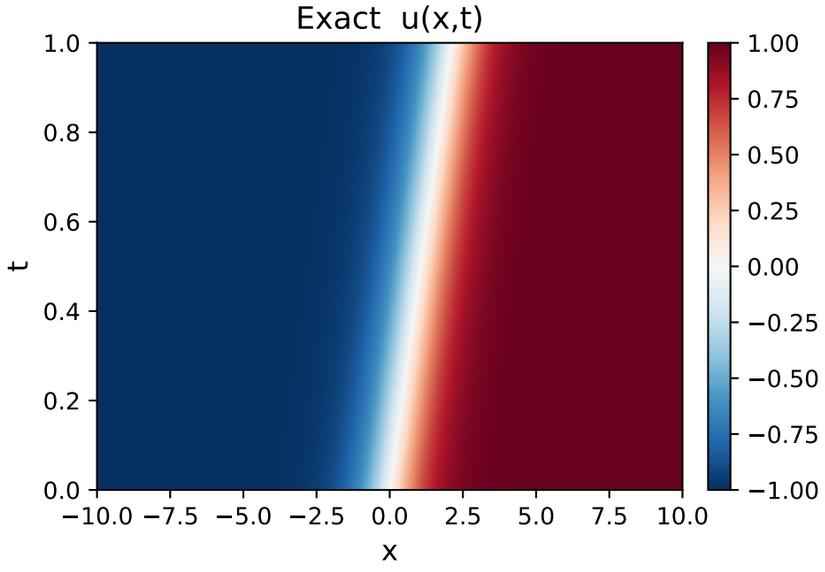
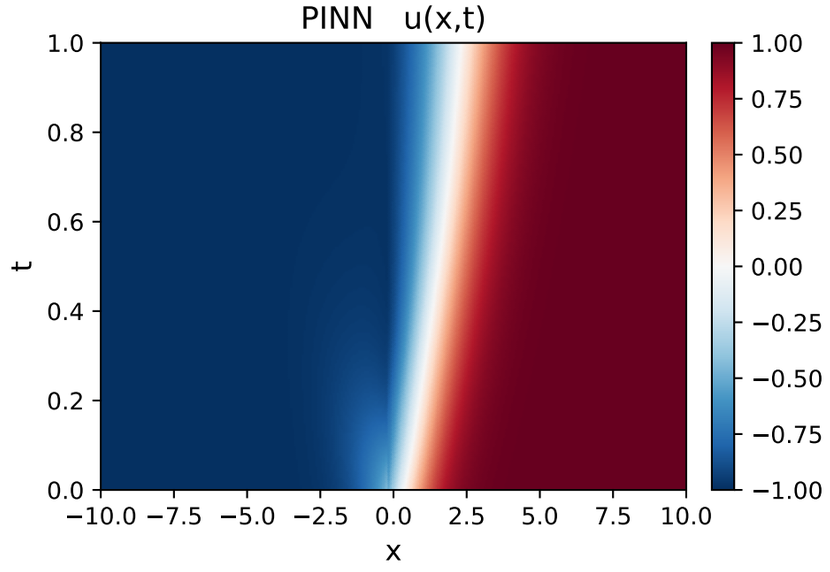
1D Allen-Cahn PINN [Adam + L-BFGS] | D_true=1.0, r_true=1.0 | D_id=4.4297, r_id=1.0033 | 5Lx64N |
n_col=2000+500kink | resample_every=50 | w=(1,10,10,0)

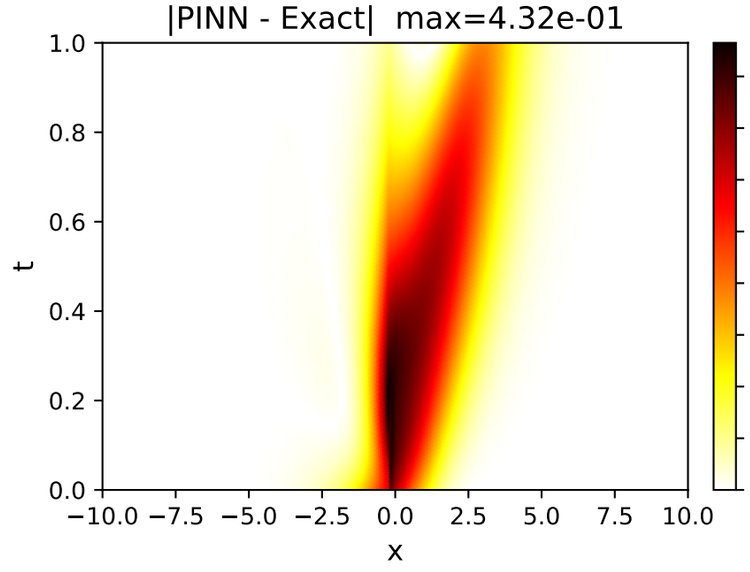
Exact u(x,t)



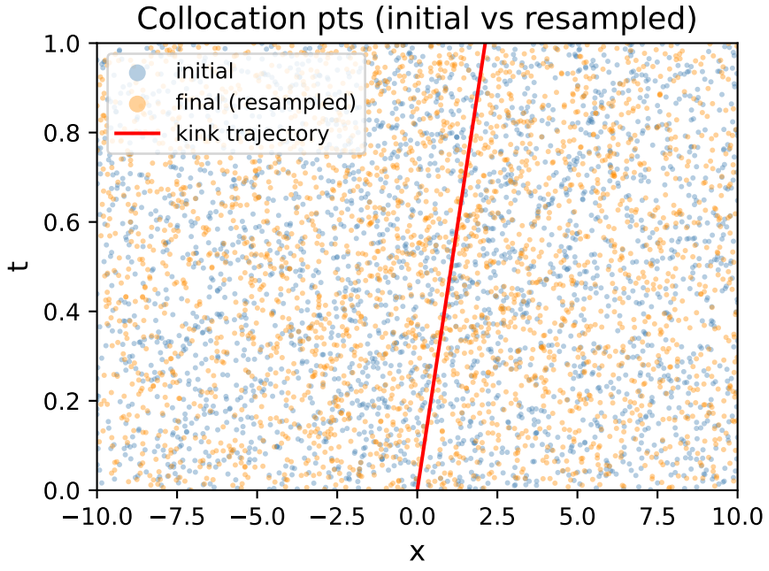
PINN u(x,t)



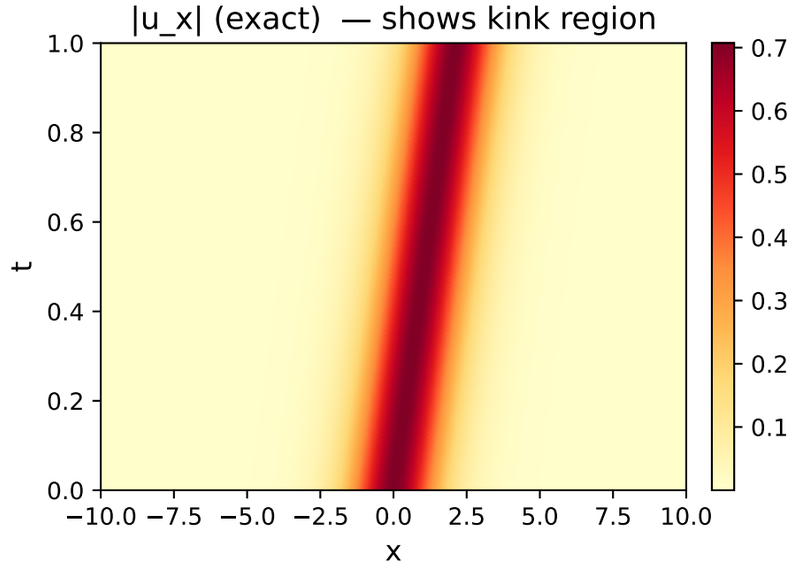
|PINN - Exact| max=4.32e-01



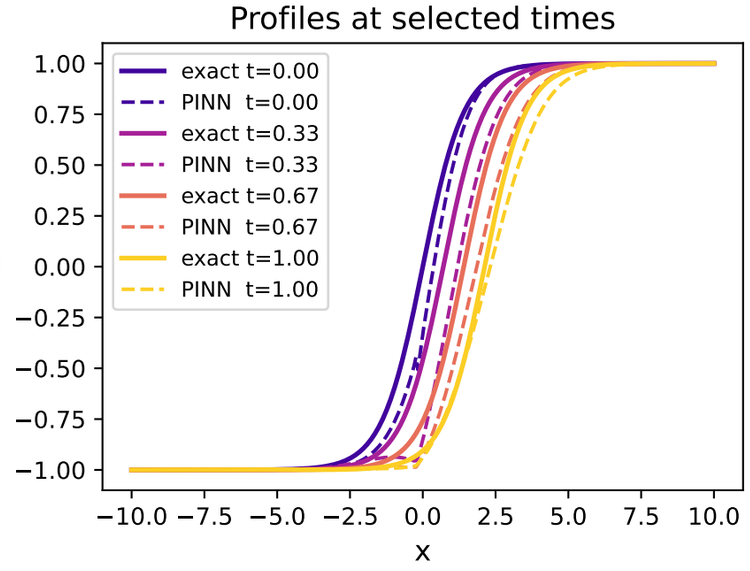
Collocation pts (initial vs resampled)



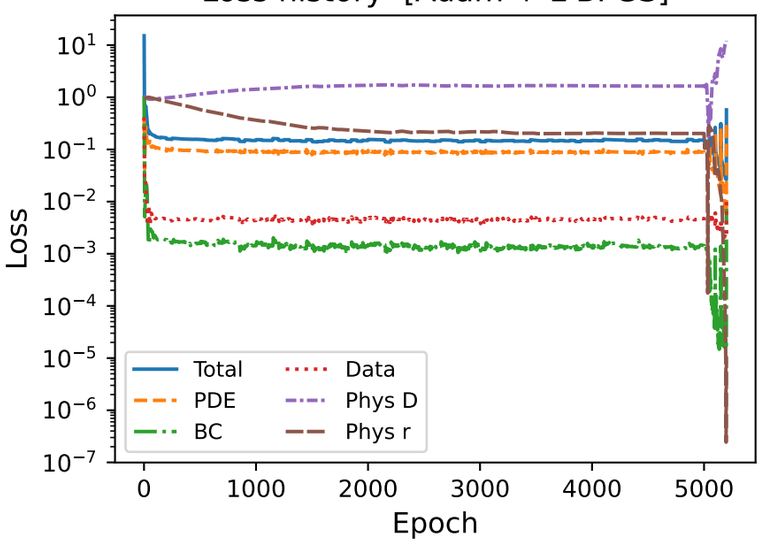
|u_x| (exact) — shows kink region



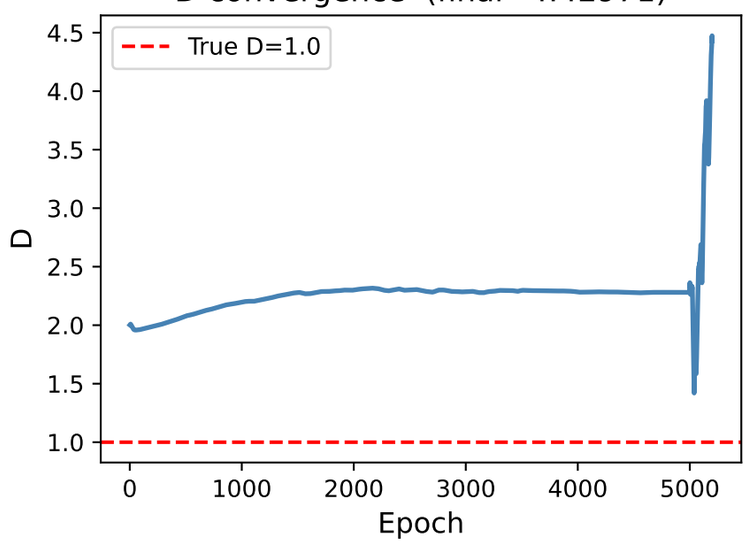
Profiles at selected times



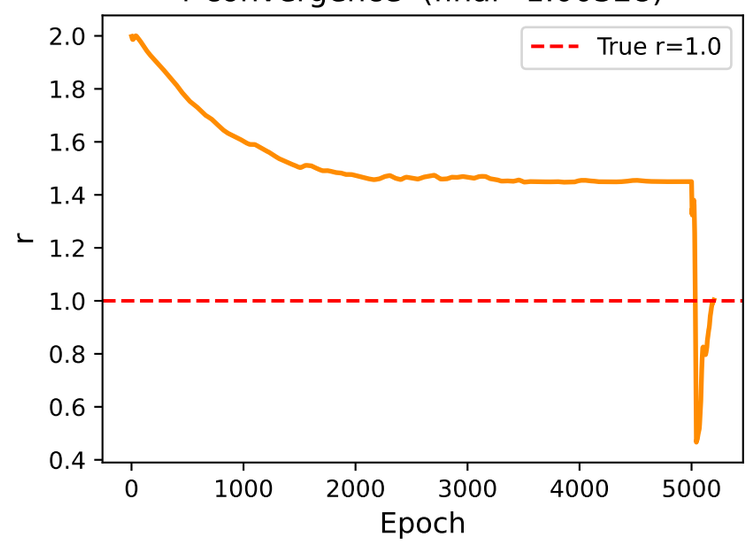
Loss history [Adam + L-BFGS]



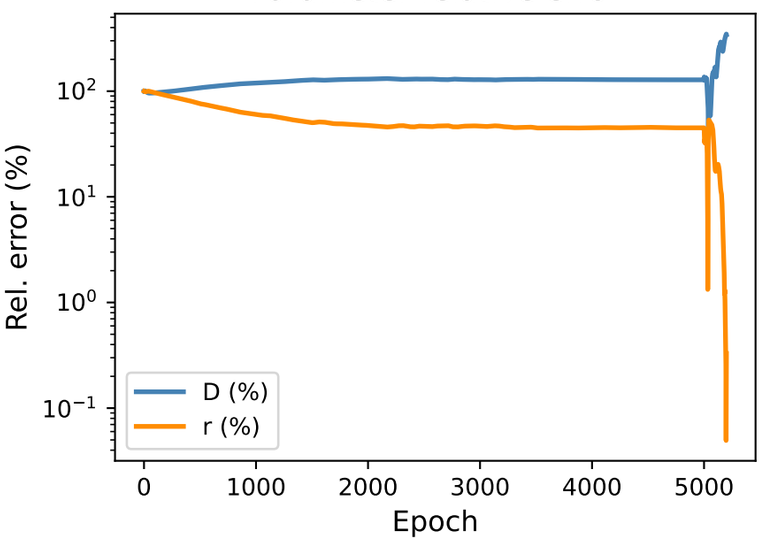
D convergence (final=4.42971)



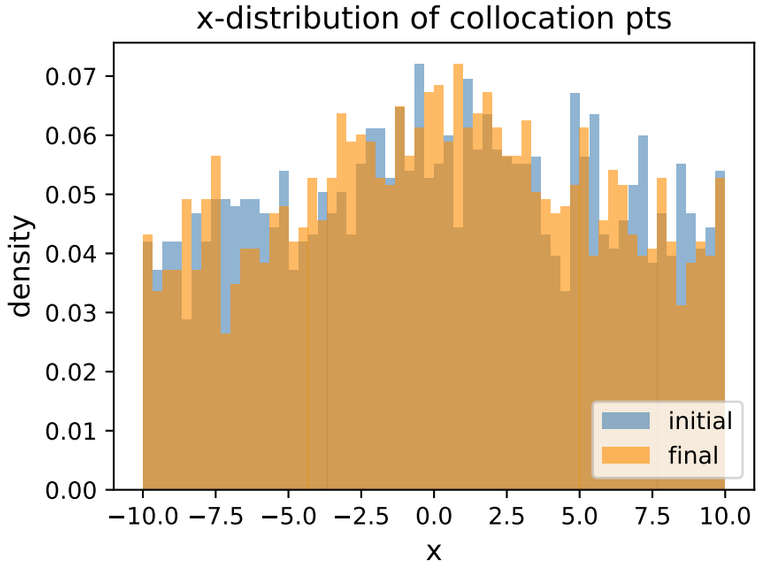
r convergence (final=1.00328)



Parameter relative error



x-distribution of collocation pts



t-distribution of collocation pts

